

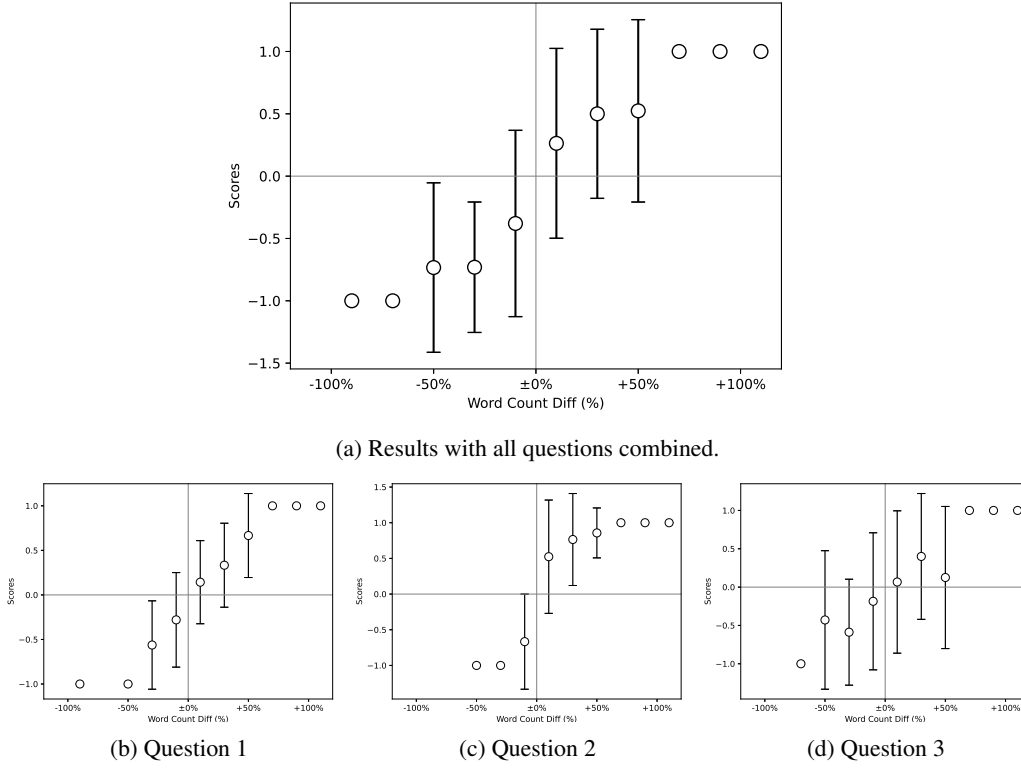


Figure 3: Example experimental results for three questions (Question 1: Blog about trip to Hawaii, Question 2: Email to professor about paper, Question 3: Blog comparing smartphones). Figure 3a combines results from all three questions and Figures 3b to 3d show results for each individual question. X-axis is the percentage of the difference between the first and the second option compared to the length of the second option. Y-axis is the actual score, with 1.0 meaning the first option was selected, 0 meaning it was a draw, and -1.0 meaning the second option was selected. There is an positive correlation between word count difference between the two options and the resulting evaluation. The data points are binned with each ranging 20%. The circles represent the average in each range and the errorbars show the standard deviation.

5 Is There A Difference in Verbosity Preference Between LLMs and Humans?

Considering that LLMs replace humans as annotators in RLAIIF, it is sufficient if LLMs could replicate human feedback and it does not necessarily have to be aligned with the ground truth. As seen in Figure 4 which plots verbosity preference of humans in the HH-RLHF dataset described later, humans seem to prefer longer answers too. Whether or not the longer answers are actually helpful is irrelevant as long as the LLM and the human come to the same conclusion. In light of this, we compare the difference in verbosity preference between LLMs and humans. We can view this as verbosity bias since the aim of LLM judgment in RLAIIF is human alignment and not the eradication of biased preference in verbosity.

We use the HH-RLHF dataset (Bai et al., 2022a) which contains human feedback data comparing pairs of answers to a prompt. It only has one feedback data per prompt, so we cannot plot the verbosity preference of humans like in the experiment in the previous chapter. Instead we can see the dissimilarity between LLMs and humans in verbosity preference in general across various questions. Precisely, this experiment looks into the relationship between the difference in a number of words in the pair of responses and the human alignment of LLMs, meaning how often LLMs give the same judgment as humans.

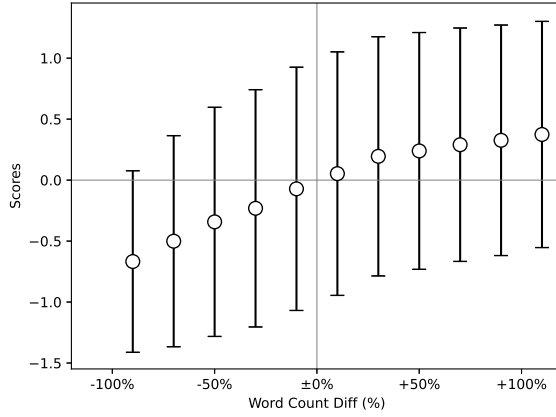
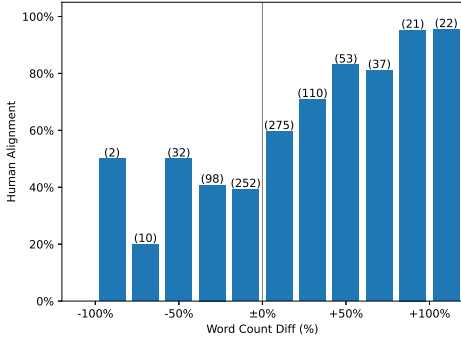
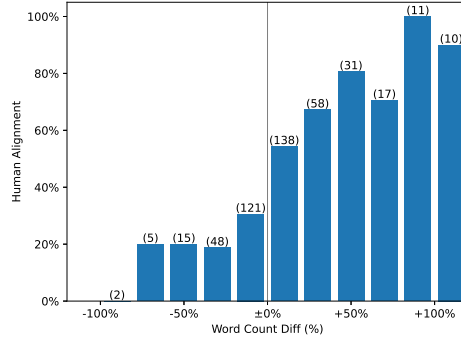


Figure 4: X-axis is the percentage of the difference between the first and the second option compared to the length of the second option. Y-axis is the resulting score by human judgement.



(a) GPT-4



(b) GPT-3.5

Figure 5: X-axis is the percentage of the difference between the chosen and the rejected option, compared to the length of the rejected option. Y-axis is the human alignment measured by the rate of LLM’s decision agreeing with humans. The numbers in brackets indicate the sample size in each bracket.

We used the same prompt template as the previous experiment but asked GPT-4 to evaluate the whole conversation. Unlike the previous experiment which evaluated answers to a single question, HH-RLHF contains conversations between a human and an assistant. Therefore we asked GPT-4 to evaluate the pair of whole conversations and answer which assistant was more helpful.

In cases where human feedback preferred the longer answer, human alignment was high for the LLMs, meaning the LLMs preferred the longer answers as well. However, when human feedback chose the answer with fewer words, human alignment was low, because the LLMs still chose the longer answers regardless of the helpfulness of the shorter answer.

One possible explanation for this is that LLMs learned to mimic human behavior heuristically by choosing longer answers – in this dataset, human feedback did tend to favor longer responses as seen in Figure 4, and it is possible the dataset used to train GPT-3.5/GPT-4 had the same tendency. Nevertheless, a closer look into the cause is up for debate.

Table 1: Verbosity bias values calculated with (6) for GPT-4 and GPT-3.5 with data from experiment.

Model	GPT-4	GPT-3.5
Verbosity Bias	0.328	0.428

6 Formulation of Verbosity Bias

In the second experiment, we observed the tendency of LLMs to have low human alignment for cases where human feedback preferred shorter answers. In this section, we formulate verbosity bias to allow for quantitative comparison between models.

In our problem setting, we define the given pair of text inputs as y_0 and y_1 , the LLM outputs decision as $Y' \in \{0, 1\}$, and the more helpful option labeled by humans as $Y \in \{0, 1\}$. We define the sensitive attribute $S \in \{0, 1\}$ which equals 0 when y_0 has more words than y_1 , and 1 when y_1 has more words than y_0 .

With these definitions, equal opportunity (Hardt et al., 2016) with respect to sensitive attribute S is satisfied if

$$P(Y' = 0|S = 0, Y = 0) = P(Y' = 0|S = 1, Y = 0). \quad (3)$$

This only accounts for cases where human feedback prefers y_0 . Although this can be attained by sorting the inputs beforehand, the equation can be generalized with accuracy parity instead of equal opportunity. Accuracy parity is satisfied if the accuracy of prediction is equal among both demographics:

$$P(Y' = Y|S = Y) = P(Y' = Y|S = 1 - Y). \quad (4)$$

The deviance from accuracy parity can be calculated with the following equation:

$$|P(Y' = Y|S = Y) - P(Y' = Y|S = 1 - Y)|. \quad (5)$$

Even though this is how the deviance is calculated in general, we thought it important that the directional information of the bias isn't lost. With the formulation below (6), a positive value indicates that the LLM prefers verbose answers, and a negative value indicates it prefers shorter answers. This distinction is crucial as some tasks may have a negative bias, for example in summarization tasks as shown in Huang et al. (2023). We also opted for the difference in *inaccuracy* between demographics

$$P(Y' = 1 - Y|S = 1 - Y) - P(Y' = 1 - Y|S = Y) \quad (6)$$

because verbosity bias refers to the inaccuracy influenced by verbosity.

Table 1 shows the verbosity bias values of GPT-3.5 and GPT-4 calculated with data from Section 5. From these numbers, we can conclude that GPT-4 has improved in verbosity bias. Compared to Wang et al. (2023), which had a limited problem setting and gave the impression that GPT-4 is significantly less prone to verbosity bias, we see that the verbosity bias still exists for GPT-4. A further experiment on other LLMs for comparison is required.

7 Discussion

7.1 Other Metrics of Equality

In the context of our study, we treat the verbosity of the response pair as the sensitive attribute in our formulation of verbosity bias in Section 6. What verbosity differs from sensitive attributes generally discussed in other cases of biases is the fact that verbosity **should** actually be taken into consideration when evaluating the responses, whereas attributes like gender or race **shouldn't** be a factor in the outcome in other cases. This is why employing other metrics of equality like demographic parity doesn't make sense here, and therefore we base the measurement of verbosity bias on equal opportunity and accuracy parity.

7.2 Limitations of Our Experiments

In the experiment in Section 4, we generate the sample responses from the same questions (before concatenation of results from all three questions). However, in our experiment in Section 5, we mix together results from various questions. This has led us to only attain the result across many kinds of questions, not the result on any specific question like in the experiment in Section 4. It is debatable which of these results is preferable.

7.3 Limitation of Our Metric of Verbosity Bias

Our formulation of verbosity bias only accounts for bias between two groups divided by whether y_0 is longer than y_1 . What it cannot detect is the bias within each of these groups; it is agnostic to the bias between cases where y_0 is barely longer than y_1 and cases where y_0 is significantly longer than y_1 . Hence, if there were to be an instance where the model has high human alignment when there is a large difference in length between the pair of responses – the plot would have a concave shape symmetrical around the vertical line down the middle – our metric would suggest that the model has close to zero verbosity bias. To avoid such a situation, showing the human alignment plot alongside the metric is recommended.

8 Conclusion

In this paper, we conducted experiments on the verbosity bias seen in LLMs’ judgment by LLMs. In previous works, the problem settings were limited and did not compare the verbosity preference to humans. With our experiments, we saw that 1) LLMs tend to favor longer answers for creative writing tasks, and 2) alignment with humans varies on verbosity with lower human alignment in cases where humans preferred shorter answers. We then formulated verbosity bias based on accuracy parity that can be used to quantitatively compare verbosity biases among models.

References

- Bai, Y., Jones, A., Ndousse, K., Askell, A., Chen, A., DasSarma, N., Drain, D., Fort, S., Ganguli, D., Henighan, T., et al. (2022a). Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*.
- Bai, Y., Kadavath, S., Kundu, S., Askell, A., Kernion, J., Jones, A., Chen, A., Goldie, A., Mirhoseini, A., McKinnon, C., et al. (2022b). Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*.
- Brants, T., Popat, A. C., Xu, P., Och, F. J., and Dean, J. (2007). Large language models in machine translation.
- Chiang, W.-L., Li, Z., Lin, Z., Sheng, Y., Wu, Z., Zhang, H., Zheng, L., Zhuang, S., Zhuang, Y., Gonzalez, J. E., Stoica, I., and Xing, E. P. (2023). Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality.
- Hardt, M., Price, E., and Srebro, N. (2016). Equality of opportunity in supervised learning. *Advances in neural information processing systems*, 29.
- Huang, K.-H., Laban, P., Fabbri, A. R., Choubey, P. K., Joty, S., Xiong, C., and Wu, C.-S. (2023). Embrace divergence for richer insights: A multi-document summarization benchmark and a case study on summarizing diverse information from news articles. *arXiv preprint arXiv:2309.09369*.
- Katz, S. (1987). Estimation of probabilities from sparse data for the language model component of a speech recognizer. *IEEE transactions on acoustics, speech, and signal processing*, 35(3):400–401.
- Lee, H., Phatale, S., Mansoor, H., Lu, K., Mesnard, T., Bishop, C., Carbune, V., and Rastogi, A. (2023). Rlaif: Scaling reinforcement learning from human feedback with ai feedback. *arXiv preprint arXiv:2309.00267*.
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. (2022). Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744.
- Stiennon, N., Ouyang, L., Wu, J., Ziegler, D., Lowe, R., Voss, C., Radford, A., Amodei, D., and Christiano, P. F. (2020). Learning to summarize with human feedback. *Advances in Neural Information Processing Systems*, 33:3008–3021.

- Wang, P., Li, L., Chen, L., Zhu, D., Lin, B., Cao, Y., Liu, Q., Liu, T., and Sui, Z. (2023). Large language models are not fair evaluators. *arXiv preprint arXiv:2305.17926*.
- Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., et al. (2023). A survey of large language models. *arXiv preprint arXiv:2303.18223*.
- Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., et al. (2023). Judging llm-as-a-judge with mt-bench and chatbot arena. *arXiv preprint arXiv:2306.05685*.